

AI Fairness & Algorithmic Accountability: An Audit of Machine Learning in Hiring

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Audit Report

Phase 1: Build and Baseline Your Model

1. For this project, I decided to choose the domain of Hiring and Employment using the Adult Income dataset from the UC Irvine Machine Learning Repository. This dataset is used to predict if an individual's income is over \$50K per year based on census data. The reason I chose this domain and dataset was because we used a very similar one in the lab so I was already familiar with the structure and variables. This would make it easier for me to focus more on fairness evaluation and policy analysis which are important parts of this project. Even though this dataset is similar to the one we used in the lab, I still chose it because this version is more biased and it would give me a good opportunity to investigate and discuss this bias in more detail. Additionally, my rationale for choosing the Hiring and Employment domain is that income level often reflects people's access to opportunities like jobs, promotions and to resources. Therefore, this domain matters because algorithmic decisions in employment and income prediction can really affect people's lives, particularly when it comes to getting a job or moving up in their career. Also, decisions in this domain usually happen early in a person's career and can have significant impact on their future. If someone is evaluated unfairly at that stage, it can limit their income and opportunities later on. I believe the people most affected are job applicants and workers, especially those from less privileged backgrounds such as women, minorities and people with low income.
2. For my EDA, I selected *income distribution*, *income distribution by gender* and *income distribution by race* as the most meaningful visualizations because they directly show class imbalance in the dataset which are important for analyzing potential bias in this project. First, when I looked at the *income distribution*, I saw that most people in the dataset earn less than or equal to \$50k while a much smaller population earns more than \$50K. This reminded me of the accuracy paradox which we learned in class, where a model can achieve high accuracy just by predicting the majority class most of the time. This also means accuracy alone can be misleading and is not enough to evaluate the model.
Next, the second graph shows *income distribution by gender*. Both males and females mostly have income less than or equal to \$50K but males have way more number of individuals earning more than \$50k compared to females and this shows that there is a sex disparity in income. From what we learned about algorithmic fairness, this matters a lot because the model may learn this pattern and reinforce it and treat sex as a signal for predicting income even if it only reflects historical bias. Finally, we looked at the *income distribution by race* graph, the difference and bias is even reflected more clearly. The dataset is heavily dominated by White individuals who also have the highest number of people earning more than \$50K. On the other hand, other racial groups have much smaller observations and this also leads to fewer individuals in the higher income group. Therefore, this imbalance could lead to the AI performing better for the majority class and worse for minority groups and result in unfair and biased outcomes between different demographic groups.
3. For this project, I trained a binary classification model using a random forest classifier to predict if an individual earns more than \$50K per year. In my preprocessing step before training the model, I first cleaned the data by removing rows with missing values using *dropna()* function. I chose this method because missing data could have a negative impact on the model's performance and introduce noise which made the model learn wrong patterns and affect predictions. Next, for feature selection, I used all available

variables in the dataset except for the target column (*income*) and the extra label column (*income_label*) I created for graphs. I decided to keep the rest because they are all relevant for predicting income such as *education*, *occupation* and *hours-per-week*. Since many of these variables are categorical, I used one hot encoding with `pd.get_dummies()` to convert them into numerical so the model can process them. Then, I split the dataset into 80% training and 20% testing, I also used `stratify=y` to make sure both dataset have a similar distribution of income population given the fact that the dataset is imbalanced. Finally, I trained a random forest classification model with 100 trees and a random state of 42. The reason I chose this model is because it can handle different types of features well and doesn't require too much tuning.

4. When it comes to the model performance, the accuracy score is approximately 0.85 and the AUC-ROC score is about 0.91 which shows the model performs generally well at predicting income. Additionally, the overall precision is 0.73, the overall recall is about 0.64 and the overall F1 score is about 0.68. Even though the overall metrics look good on average, this doesn't suggest good performance and there might be some differences between subgroups so I also analyze disaggregated results by the protected attribute *sex*. When breaking the disaggregated metrics down, there are some differences between females and males. Specifically, for females, the model achieves higher accuracy of approximate 0.92 but lower recall score of around 0.52. On the other hand, for males, the accuracy score is lower of around 0.82 but recall score is higher of around 0.65. This one is important because based on what we learned in lectures, recall is the metric that shows how well the model predicts true positive rate or individuals who actually earn more than \$50K per year in this case. Therefore, this suggests that the model is better at predicting males that have income of over \$50K compared to females. In other words, it has worse performance on predicting women with higher income. Because the disaggregated metrics show the model performs differently between subgroups, this suggests potential bias in the dataset which is then learned by the AI and leads to different prediction performance across sex.

Before moving to Phase 2: Before moving to phase 2, I expect bias to appear mostly across sex and race in my model. From the EDA, I already noticed that males are more likely to be in the >\$50K income group compared to females. Furthermore, the dataset also has more White individuals compared to other races, as shown in the *income distribution by race* graph and there are also many more males observations in the dataset, as shown in the *sex distribution* visualization. Because of these patterns, I believe the model will learn these existing imbalances and reflect them in its predictions. To be specific, as mentioned in part 4, I expect the model to perform better for males when predicting individuals with annual income more than \$50K which already showed up in higher recall and precision score compared to females. I also think race could be another area where bias appears, since compared to White, other racial groups are underrepresented in the dataset. With fewer data points and observations for women and other racial groups, the model might not learn those patterns as well which could lead to worse performance for these populations.

Phase 2: Fairness Audit and Explainability Analysis

Fairness Analysis

1. For my fairness analysis, I computed all metrics separately for each protected group rather than using overall averages so I could better identify disparities across sex. Starting with demographic parity or statistical parity, I looked at the positive rates or the percentage of people predicted to have annual income of over \$50K. The results show some disparities

between two gender groups with approximately 27% of males are predicted to earn above \$50K annually, compared to only about 7.98% of females. Next, for equalize odds or equal opportunity, I checked if the model correctly predicts people who have over \$50K in annual income at the same rate. In other words, I calculated recall or true positive rate for each gender group. The results also show that males perform better of around 65%, compared to females of approximately 52%. Finally, for predictive parity, I calculated precision for each group. This fairness metric is relatively similar between two gender groups, with values of 0.737 for males and 0.718 for females.

2. In my results, the fairness metrics conflict with each other because each metric focuses on a different fairness goal. For instance, for demographic parity, this fairness metric focuses on equal probability of success between groups and my model clearly fails this because it predicts males as having annual income of over \$50K at a much higher rate than females, 27.04% compared to 7.98%. This data shows that the AI assigns positive predictions much more frequently to males. On the other hand, for predictive parity which is the fairness metric that focuses on whether positive predictions are equally qualified, the data looks much more balanced with 0.737 for males and 0.718 for females. These numbers show that it is similarly accurate for both groups when the AI predicts an individual as having annual income of over \$50K. The conflict happens because the model performs generally well with predictive parity but can look unfair when evaluated using demographic parity. Additionally, the equalized odds metric also shows inequality with males recall at 0.646 and female recall at 0.515, meaning the model is less successful at predicting individuals with annual income over \$50K for females. The conflict appears again here because improving equal opportunity or recall for females would likely require increasing positive predictions for this gender group which could also reduce predictive parity or precision.

Based on this, I would prioritize equal opportunity. In my data results, recall or true positive rate for males is 0.646 while for females it is only 0.515. These numbers suggest that among people who actually earn over \$50K per year, the model is less likely to correctly predict females. To put it simply, qualified females are being missed more often by the model or higher false negatives, which can be a significant problem and directly limit their access to opportunities like hiring or promotions. This choice connects to justice AI ethics or fair ethics which is the ethical framework focused on fairness and avoiding discrimination in decision-making. Based on the results, the model is creating a situation where one group is being treated unfairly even when they are equally qualified. When it comes to justice ethics, this is concerning because the goal is to reduce discrimination and make sure all groups are treated equally. Therefore, by prioritizing equal opportunity, my goal is to focus on making sure that qualified individuals in both genders are treated the same by the model. At the same time, I also know there is a trade-off of justice ethics in AI. Specifically, if I try to increase equal opportunity or recall for females at 0.515, I would also likely need to increase predictions of individuals who earn more than \$50K per year for this group so the number would be closer to males at 0.646. But it could also lower precision or predictive parity because some of those added predictions might be wrong and lead to more false positives. Moreover, it could change demographic parity since the current rates are 7.98% for females and 27.04% for males. Therefore, there is mathematical tension in my results as improving equal opportunity may reduce predictive parity or changing selection rates between groups. I also think prioritizing equal opportunity may reduce the performance of the AI and it could be less accurate. But I still believe that missing qualified people is a more serious problem than having perfect metrics and numbers so I would choose to prioritize equal opportunity even though there are trade-offs.

Explainability Analysis

1. For this part, I used SHAP to generate global feature importance explanations using the SHAP beeswarm summary plot. Based on this plot, the top 5 features that drive the model in predicting income are *marital-status*, *age*, *capital-gain*, *occupation* and *education-num*. These variables have the largest spread of SHAP values which means they matter the most on the predictions of the model. In this dataset, the main protected attribute I used is *sex* but other protected attributes such as *race* and *age* are also relevant here. Even if they are not directly used, the model can still rely on proxy variables. For example, *occupation* can act as a proxy for both gender and race because some certain roles have historically been filled by specific demographic groups. As a result, I think the model might associate those occupations to higher or lower income and this is problematic because even if the model does not directly use gender or race, it can still indirectly guess them through occupation. Furthermore, the model would be relying on existing social inequalities which can result in biased and unfair predictions. Similarly, *marital-status* can be a proxy for gender. In real life, income patterns often look different depending on their marital status and I think these differences are tied to gender roles not just their skills or ability. For example, married men usually have higher income than other groups. Therefore, when the model uses *marital-status* as one of the most influential features in predicting income, it could be picking up on hidden gender patterns and this is problematic since there might be bias in its predictions. For example, the model might predict that a married individual is more likely to earn over \$50K per year.
2. For the Recourse Test, I selected 1 individual who received an unfavorable prediction of having an annual income less than \$50K. Based on the decoding information of this person, I know that he is 23 years old, has a high school education, works in a craft-repair job, works 37 hours per week and has no capital gain. Based on the SHAP waterfall plot, the strongest factors pushing the prediction toward $\leq 50K$ are *marital-status* and *occupation*. Other factors with smaller effect are *capital-gain* and *relationship*, followed by *sex* and *hours-per-week*. If I were this person, I would find this result discouraging because some of these factors are not easy to change or actionable. For example, marital status is a personal life choice and not something someone should be expected to change just to receive a favorable outcome. The same applies to other factors such as *relationship* or *sex* which are either personal or not changeable at all. However, there are some factors that are more actionable for this person to receive a favorable outcome. Specifically, this person would likely need to switch to a job that pays more, increase their work hours per week and build up savings or investments to improve their *capital-gain* feature. These changes are actionable but they take time, effort and resources so they cannot be achieved quickly. Also, even though the SHAP waterfall plot shows which factors influenced the decision, a non-technical person could not easily understand or act on. The model shows important variables but does not clearly translate them into specific and exact steps. According to GDPR Article 22's right to explanation, organizations need to explain automated decisions. In this case, the explanation is not enough and needs more details because it only shows key features but does not tell why it made that decision and offers clear or actionable recommendations on what to do next. **Discovery Checkpoint:** Based on my predictions in Phase 1, I thought bias would mostly appear in *sex* and *race* since I already observed some imbalance in the dataset shown up during exploratory data analysis. After running the random forest model and checking for fairness metrics among protected groups, I think my predictions were partially right. All numbers showed that the model performed better for males compared to females in predicting individuals with annual

income above \$50K which indicated the presence of gender bias. This made sense based on my initial expectation since males were both having more observations in the dataset and more likely to receive favorable predictions by the model. In contrast, what surprised me though was that bias did not just come from obvious features like *sex* because when I looked at the SHAP beeswarm summary and waterfall plot, I noticed that features such as *marital-status* and *occupation* had an important influence on predictions. Even though these variables did not seem to be related directly to bias but they still can act as proxy variables and pick up on hidden patterns so even if we do not directly use *sex*, the model can still reflect gender bias in real life through these variables.

Phase 3: Regulatory and Policy Mapping

1. For this project, I mapped my technical findings to some regulations that are relevant to the hiring and employment domain. In this domain, automated systems can influence decisions like hiring or promotion. First, I decided to use an international regulation or GDPR Article 22 from EUR-Lex which gives individuals “the right not to be subject to decisions based solely on automated processing”. I also used GDPR Article 13 which requires providing “meaningful information about the logic involved.” These articles connect to my Recourse Test in Phase 2 because I believe people that are affected by these systems should be able to understand what factors influenced the decision and what they can do to improve the outcome. I also included Article 5(1) of GDPR which emphasizes fairness and transparency in data processing because systems in this domain must not only be accurate but also fair and understandable to the people affected by them. Also, for the U.S. guideline, I mapped my findings to the NIST AI Risk Management Framework (AI RMF 1.0), specifically Section 3 AI Risks and Trustworthiness and Section 3.7 Fair - with Harmful Bias Managed. Section 3 mentions that trustworthy AI systems must be accountable, transparent and fair. In addition, Section 3.7 focuses on identifying and managing different types of bias. These sections are related to my findings since there are some potential biases in both my model’s data and its predictions. Furthermore, I also used Section 5.3 Measure or MEASURE 2.11 in this section which requires evaluating fairness and bias and documenting the results and this is what I did in my fairness analysis by comparing equal opportunity and predictive parity among protected groups. Also, as stated in page 1 of the NIST framework, “AI systems can amplify, perpetuate, or exacerbate inequitable or undesirable outcomes” if risks are not controlled, this also related to the bias I found in my model and dataset. Finally, according to Title VII of the Civil Rights Act of 1964, Section 703, it is illegal for employers to discriminate based on race, sex or other protected attributes. I believe this section is directly related to my findings since my AI model showed differences in performance among sex which means similar systems in hiring could lead to biased decisions.

2. Compliance table:

Findings in Phase 2	What the system does	Regulation and section	Compliance and what the regulation requires	Reasoned explanations

Gender bias in demographic parity, 27% vs. 7.98%.	The model predicts males as having annual income over \$50K much more often than females.	Section 703, Title VII of the Civil Rights Act of 1964	Non-compliant. This section requires that decisions are not discriminatory or biased based on sex.	There is a significant difference between 2 numbers which suggests potential bias since females are selected at a much lower rate. In a hiring context, this could be seen as unfair.
Difference in equal opportunity or recall between two gender groups with females at 52% and males at 65%.	The model misses more qualified females. In other words, it has more false negatives or incorrectly predicts more females as earning <=\$50K per year.	Section 703, Title VII of the Civil Rights Act of 1964	Non-compliant. This section requires equal treatment of protected groups.	Because the data shows lower recall or true positive rate for females, this means in hiring, qualified female candidates are more likely not to be chosen and this is not fair.
Predictive parity	The model	Section 3.7,	Gray area. This	At first the

is similar between gender groups.	shows close precision or predictive parity across genders, 0.737 vs. 0.718.	NIST AI RMF 1.0	section states that bias should be minimized in the system.	results look fair but when I observed data in other fairness metrics, there are still clear differences between males and females.
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Conflict happens between fairness metrics.	Based on the calculations, I saw that the fairness metrics conflict with each other and could not be satisfied at the same time	Section 3, NIST AI RMF 1.0	Compliant. This section mentions that negative AI risks must be identified and managed in order for the system to be trustworthy.	I was able to explain the trade-offs associated with the fairness metric I chose to prioritize and this matches the NIST framework that AI risks need to be identified and understood.
SHAP explanations through beeswarm summary and waterfall plots.	Through SHAP explanations, the model shows that features like marital-status, occupation and capital-gain matter the most in its predictions.	Article 22 and Article 13, GDPR	Gray area. Article 22 states that people shouldn't be subject to decisions made only by automated systems and they should be able to challenge those decisions. Article 13 also requires providing "meaningful information about the logic involved."	In this case, the model shows which features matter the most through SHAP explanations but it is difficult for a non-technical person to understand. It also does not clearly explain how that person can respond to or challenge the decision.
The recourse given to the person who gets the unfavorable prediction is not very actionable.	The system includes a SHAP waterfall plot showing feature impact on its predictions.	Article 22 and Article 13, GDPR	Non-compliant. Article 22 says people can challenge automated decisions and Article 13 requires clear explanations of	The model shows which features affect the prediction but it does not give realistic actions the person can take to change the

			how those decisions are made.	outcome. For example, changing marital status or relationship is not reasonable so it does not support the user to respond or contest the decision.
The model relies on features such as occupation and marital-status which are proxies for protected attributes.	The system uses these features as strong variables in the model.	Section 3.7, NIST AI RMF 1.0	Non-compliant. This section requires that the system must identify and manage different types of bias and avoid reinforcing existing inequalities.	The features the model relies on can act as proxy variables for protected attributes like gender which can lead to indirect bias in the predictions of the model.
Bias is identified through fairness metrics.	I used the system to calculate and compare fairness metrics between protected groups.	Section 5.3, Measure 2.11 NIST AI RMF 1.0	Compliant. This section requires AI risks or fairness and bias should be evaluated and documented	This meets the requirement to evaluate and document bias because I used fairness metrics to check for bias and compare results between groups.
The model gives different results depending on the fairness metrics used, demographic parity, equal opportunity, and predictive parity.	The system shows inconsistent results between different fairness metrics.	Section 3, NIST AI RMF 1.0	Gray area. Section 3 says that organizations need to manage negative risks in AI systems.	The model clearly shows fairness trade-offs but I believe the risk is recognized but not really managed.

Based on the SHAP waterfall plot, the prediction of the model is influenced by	The system uses the SHAP waterfall plot to show which features have the biggest	Article 13, GDPR	Gray area. Article 13 requires providing meaningful information	Even though the model shows the contribution of each feature to the prediction, I
multiple features.	impact on the prediction.		about the logic involved in automated decisions and their consequences for the person.	think it is still hard for a non-technical person to fully understand the decision. The information is not really meaningful in practice since the person cannot clearly understand the logic and the consequences of the outcome.

3. For regulatory comparison, I focus on the explainability gap from my audit. Even though SHAP outputs are provided, I believe it is still difficult for a person with no technical background to understand. In the first place, it looks like all three regulations and frameworks are aligned since they all emphasize the importance of transparency but when I look closely, they actually approach this issue in very different ways. To be more specific, as mentioned above, under Article 13 of GDPR, there is a strong focus on giving people “meaningful information about the logic involved” and this basically means the explanation should be clear, easy to understand and actually useful for the person affected by the decision. In contrast, Title VII of the Civil Rights Act of 1964 in the U.S. does not directly require explainability since it focuses more on whether the outcomes are discriminatory and not on how the decisions are explained. On the other hand, under Section 3 of NIST AI Risk Management Framework, explainability is treated as one of many characteristics of trustworthy AI that organizations should balance, this is a best practice and not a strict requirement. Therefore, all three frameworks agree that fairness and transparency are all important but they diverge in enforcement. Additionally, I think the non-obvious insight here is that a system like mine could technically pass under U.S. law if it avoids discrimination outcomes and also seems fine under NIST guidelines but I think it still fails under General Data Protection Regulation because the explanation is not actually useful or understandable for the person. Based on this, it leads to deeper conflict. GDPR treats explanation as a user right while Title VII of the Civil Rights Act of 1964 focuses more on fairness and NIST treats both as flexible design goals. Because of this, I believe that what falls through the cracks is the usability of explanations or whether the explanation is actually useful, particularly my model technically gives explanations and transparency but they are not actually helpful for real users.

Phase 4

Pre-Deployment Checklist:

1. Before deploying the predictive model, the organization must calculate fairness metrics such as demographic parity, equal opportunity and predictive parity between gender groups and document any disparities and biases since my model showed a large difference of 27% vs. 7.98% in positive prediction rates or demographic parity.
2. The organization should set a clear threshold for how much difference in recall between males and females and retrain the model if the gap exceeds that threshold because my model showed lower recall for females of 0.515 vs. 0.646 for males.
3. The model should be checked for proxy bias by looking at whether features like *occupation* and *marital-status* are acting as indirect indicators of gender. If they are introducing bias, they should be removed or adjusted, linked to Section 3.7 of NIST on managing bias.
4. The organization should clearly document any fairness trade offs before deployment like equal opportunity versus predictive parity and explain why one metric is prioritized over the other, this matters because my results showed that these metrics can conflict with each other. This also aligned with Section 3 of NIST on managing negative AI risks.
5. The organization should include a human review process in the system for people who receive unfavorable predictions so they have a way to contest the automated decisions and meet the requirements in General Data Protection Regulation Article 22 and Article 13.
6. Instead of just showing technical outputs like SHAP plots, the organization should provide a simple explanation in plain English of the main reasons behind each decision so a normal person can understand why they got that result to meet GDPR Article 13's requirement for meaningful information.
7. The organization should translate model and system outputs into clear and actionable recourse steps, like increasing work hours or improving skills and also clearly separate which factors can be changed and which cannot since my Recourse Test shows that some factors like *marital-status* and *sex* are not actionable.
8. The organization should check whether the suggested recourse actions are actually realistic and achievable within a reasonable timeframe since my audit showed some factors like *marital-status* are not really actionable.
9. The model should be evaluated on a dataset that has more balanced observations between gender groups since my audit in Phase 1 EDA showed that the dataset is imbalanced and could lead to some biases in predictions.
10. The organization must document all fairness metrics, explanation methods and recourse strategies in a formal report before deployment and this follows Section 5.3 of NIST AI Risk Management Framework which focuses on measuring and documenting risks.

Stakeholder Disclosure Document

It is important that people affected by this model understand how it works and how decisions are made so the goal of this document is to explain how the system works in a simple and clear language. Since my audit and model involve a lot of technical details and are not really explainable, it can be hard for people to understand and they can find the system unfair and unreasonable. Therefore, this stakeholder disclosure document is meant to break everything down in a clear and simple way, including what the model does, what data it uses to make decisions, how it was evaluated for fairness, and what users can do if they disagree with a decision.

First, in terms of what the model does, I built an AI machine learning model that predicts whether someone is likely to earn more than \$50,000 per year based on their background information and this type of prediction can be used to support decisions related to hiring or promotions. That said, this system is not perfect and should not be used as the only factor in making decisions. A human reviewer can step in if people have concerns or questions.

Next, when it comes to the data the system uses to make predictions, it relies on general background information from individuals such as:

- age
- education level
- type of job
- working hours
- financial-related information like capital gain, savings or investment income

The system can also use other related information from the dataset to make predictions. Some of this information had a stronger influence on the prediction, especially job type, marital status and education.

Additionally, I also tested the system to make sure it treats people fairly between different groups. For example, I checked whether men and women are treated equally and I also checked if the system correctly identifies people who should receive better outcomes.

I found that the system performs better for some groups than others particularly between males and females. Because of this, I believe there are still several areas where the model can be improved to make it fairer and reduce these biases.

Therefore, if the affected individuals believe their results are incorrect or unfair, they have the ability to ask for a human review of their case, request a clearer and more detailed explanation of their result and also provide additional information that may change and improve their outcome. This is really important because my audit showed that the model does not always give clear or actionable explanations.

I also feel that this matters a lot since the model is based on historical data which can include existing bias. As a result, it cannot fully reflect a person's true abilities or potential and the outcome could be biased. For this reason, results from this model should always be used carefully and reviewed by human experts the model should only be used as a support tool.

This project was developed in Google Colab using Python 3.x. All required libraries are listed in the requirements.txt file.

Google Colab link:  IS467 Final Project - thuyn3

Phase 1: Build and Baseline Your Model

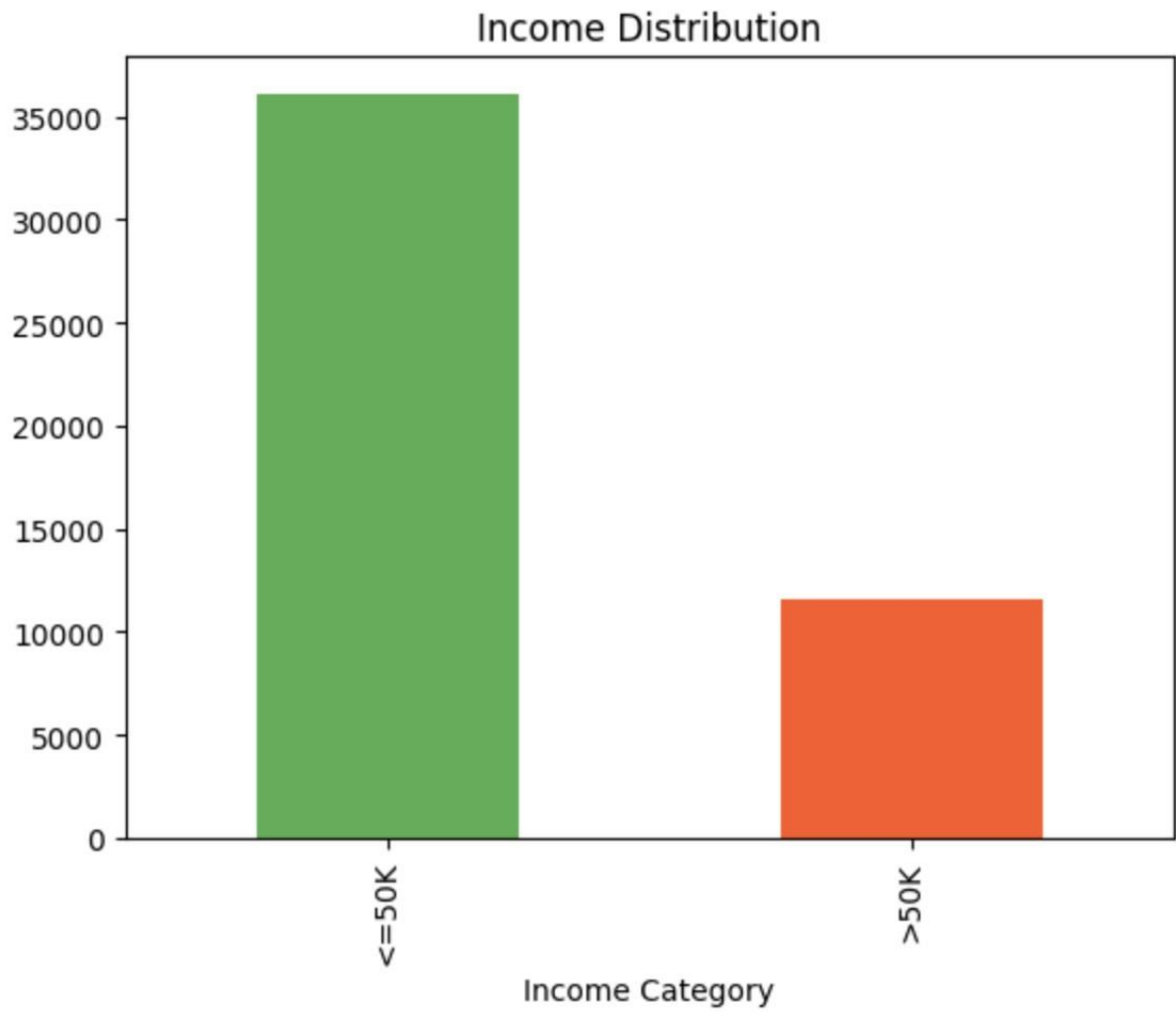
```
# Import required libraries
!pip install shap lime folktables -q
import seaborn as sns import numpy
as np import pandas as pd import
matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split from
sklearn.preprocessing import LabelEncoder from sklearn.ensemble
import RandomForestClassifier from sklearn.metrics import (
classification_report, accuracy_score, ConfusionMatrixDisplay
)
import shap
!pip install ucimlrepo

# Import the dataset
from ucimlrepo import fetch_ucirepo
# fetch dataset
adult = fetch_ucirepo(id=2)
# data (as pandas dataframes) X =
adult.data.features y =
adult.data.targets # metadata
print(adult.metadata) # variable
information print(adult.variables)

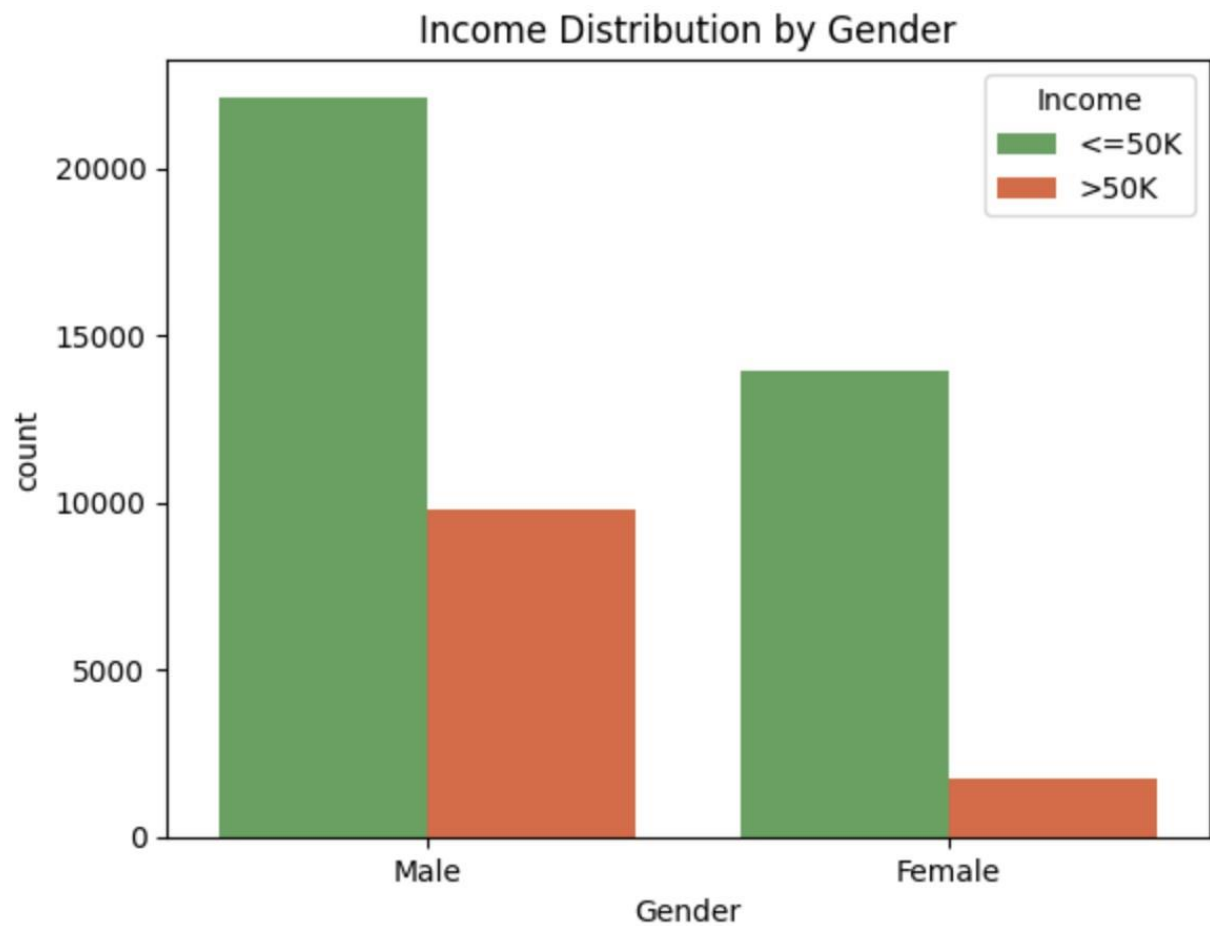
# Combine everything into one dataframe for easier processing
import pandas as pd df = pd.concat([X, y], axis=1) # Preview first 5
rows of the dataset df.head()

# Drop missing values in the dataset df =
df.dropna()
# Convert income column into binary (1 = >50K, 0 = <=50K)
df["income"] = df["income"].apply(lambda x: 1 if ">50K" in str(x) else 0)
# Check dataset structure
df.info()

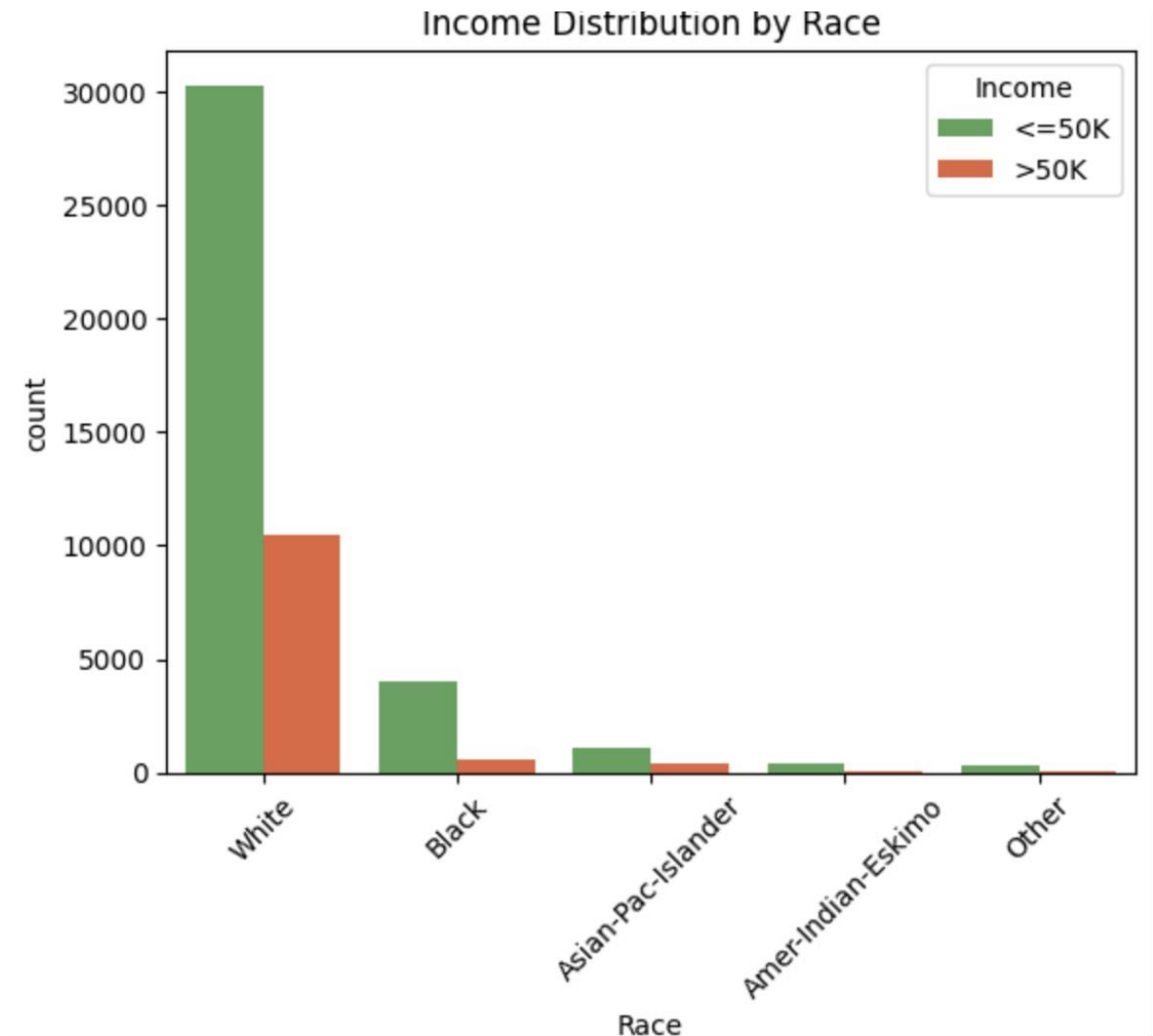
# EDA
# Create a readable label column for income df["income_label"] =
df["income"].map({0: "<=50K", 1: ">50K"})
# Income distribution
df["income_label"].value_counts().plot(kind="bar", color=["#4CAF50", "#FF5722"])
plt.title("Income Distribution") plt.xlabel("Income Category") plt.show()
```



```
# Income distribution by gender
sns.countplot(data=df,x="sex",hue="income_label",palette=["#4CAF50", "#FF5722"])
plt.title("Income Distribution by Gender") plt.xlabel("Gender") plt.legend(title="Income")
plt.show()
```



```
# Income distribution by race
sns.countplot(data=df,x="race",hue="income_label",palette=["#4CAF50", "#FF5722"])
plt.title("Income Distribution by Race") plt.xlabel("Race") plt.xticks(rotation=45)
plt.legend(title="Income") plt.show()
```



```
# Sex distribution df["sex"].value_counts().plot(kind="bar",
color=["#4CAF50", "#FF5722"]) plt.title("Sex Distribution") plt.xlabel("Sex")
plt.ylabel("Count") plt.show() # Race distribution
df["race"].value_counts().plot(kind="bar", color=["#4CAF50", "#FF5722"])
plt.title("Race Distribution") plt.xlabel("Race") plt.ylabel("Count")
plt.xticks(rotation=30) plt.show()
# Education distribution df["education"].value_counts().head(6).plot(kind="bar",
color=["#4CAF50", "#FF5722"]) plt.title("Education Level") plt.xlabel("Education")
plt.ylabel("Count") plt.xticks(rotation=30)
plt.show()
```

```
# Preprocessing # Clean
missing values df_clean =
df.dropna().copy()
```

```
# Feature selection #
Target variable y =
df_clean["income"]
# Drop the target column from the features
```

```

X = df_clean.drop(columns=["income", "income_label"], errors="ignore")
# One-hot encoding for categorical variables
X_encoded = pd.get_dummies(X, drop_first=True)

# Train/test split
# Split dataset into 80% training and 20% testing
X_train, X_test, y_train, y_test = train_test_split( X_encoded,
    y,
    test_size=0.20,
    random_state=42, stratify=y
)

# Train Random Forest model rf =
RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
# Fit the model using the training data rf.fit(X_train,
y_train)

# Predictions y_pred =
rf.predict(X_test)
y_prob = rf.predict_proba(X_test)[:,-1]

# Import metrics from sklearn.metrics import accuracy_score, precision_score, recall_score,
f1_score, roc_auc_score # Calculate metrics accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred) recall = recall_score(y_test, y_pred) f1 =
f1_score(y_test, y_pred) auc = roc_auc_score(y_test, y_prob)
# Print results print("Accuracy:",
accuracy) print("Precision:",
precision) print("Recall:", recall)
print("F1 Score:", f1) print("AUC-
ROC:", auc)

# Disaggregated # Create a
test dataframe df_test =
X_test.copy()
df_test["actual"] = y_test
df_test["pred"] = y_pred
df_test["prob"] = y_prob
# Add gender column back from original dataframe
df_test["sex"] = df.loc[X_test.index, "sex"] # Function
to compute metrics def compute_metrics(group):
    return pd.Series( {
        "accuracy": accuracy_score(group["actual"], group["pred"]),      "precision":
precision_score(group["actual"], group["pred"]),
        "recall": recall_score(group["actual"], group["pred"]),
        "f1": f1_score(group["actual"], group["pred"]),
    })

```

```

        "auc": roc_auc_score(group["actual"], group["prob"])
    })
# Apply the function to each gender group metrics_by_gender =
df_test.groupby("sex").apply(compute_metrics) print(metrics_by_gender)

```

Phase 2: Fairness Audit and Explainability Analysis

Fairness Analysis

```

# Add group columns df_test =
X_test.copy()
df_test["actual"] = y_test
df_test["pred"] = y_pred
df_test["prob"] = y_prob
# Add protected attribute gender column df_test["sex"] =
df_clean.loc[X_test.index, "sex"]

```

```

# Demographic parity dp
= df_test.groupby("sex")["pred"].mean()
print("Demographic Parity:\n", dp)

```

```

# Equal opportunity for gender in
df_test["sex"].unique(): a =
df_test[df_test["sex"] == gender] b =
recall_score(a["actual"], a["pred"])
print(f'{gender} Equal Opportunity:', b)

```

```

# Predictive parity
for gender in df_test["sex"].unique(): a =
df_test[df_test["sex"] == gender] c =
precision_score(a["actual"], a["pred"])
print(f'{gender} Predictive Parity:', c)

```

Explainability Analysis

```

# Take a smaller sample
X_sample = X_test.sample(300, random_state=42)
# Create SHAP explainer for the model
explainer = shap.TreeExplainer(rf) #
Calculate SHAP values for test data
shap_values = explainer.shap_values(X_sample)

```

```

# Handle shape if
isinstance(shap_values, list):
shap_class_1 = shap_values[1] elif
len(shap_values.shape) == 3:
shap_class_1 = shap_values[:, :, 1] else:
shap_class_1 = shap_values

```

```

# Create mapping from encoded to original feature
feature_map = {} for col in X_sample.columns: if
"_" in col:
    feature_map[col] = col.split("_")[0] else:
    feature_map[col] = col # Convert to DataFrame shap_df =
pd.DataFrame(shap_class_1, columns=X_sample.columns)
# Group SHAP values by original feature shap_grouped =
shap_df.groupby(feature_map, axis=1).sum()
# Also group input features
X_grouped = X_sample.copy()
X_grouped.columns = [feature_map[col] for col in X_sample.columns]
X_grouped = X_grouped.groupby(level=0, axis=1).mean()

# Beeswarm SHAP summary plot
shap.summary_plot(shap_grouped.values, X_grouped, feature_names=X_grouped.columns)

# Find one unfavorable prediction, pred = 0 means <=50K pred_low =
df_test[df_test["pred"] == 0].index[0]

# Show that person's information, decoded X_test_reset =
X_test.reset_index(drop=True) y_test_reset =
y_test.reset_index(drop=True) original_test =
df_clean.loc[X_test.index].reset_index(drop=True) pred_low =
np.where(y_pred == 0)[0][0] print("=" * 60) print(f'Sample
#{pred_low} — Predicted: <=50K") print("=" * 60) for col in
original_test.columns:
    if col not in ["income", "income_label"]:
        print(f'{col:<20} {original_test.loc[pred_low, col]}') print("\nPrediction:",
"<=50K" if y_pred[pred_low] == 0 else ">50K") print("Actual: ", "<=50K" if
y_test_reset.iloc[pred_low] == 0 else ">50K")

# Original feature names original_features = [
    "age", "workclass", "education", "education-num",
    "marital-status", "occupation", "relationship",
    "race", "sex", "hours-per-week", "native-country"] # Map
encoded columns back to original feature groups
feature_map = {} for col in X_test.columns:
    if col.startswith("workclass_"):
        feature_map[col] = "workclass" elif
col.startswith("education_"):
        feature_map[col] = "education" elif
col.startswith("marital-status_"):
        feature_map[col] = "marital_status" elif
col.startswith("occupation_"):
        feature_map[col] = "occupation" elif
col.startswith("relationship_"):
        feature_map[col] = "relationship" elif

```

```

col.startswith("race_"):
feature_map[col] = "race"    elif
col.startswith("sex_"):
feature_map[col] = "sex"    elif
col.startswith("native-country_"):
feature_map[col] = "native_region"    else:
    feature_map[col] = col

# Convert SHAP values into a dataframe shap_df =
pd.DataFrame(shap_class_1, columns=X_test.columns)
# Group SHAP values by original feature shap_grouped =
shap_df.groupby(feature_map, axis=1).sum()
# Group feature values too
X_grouped = X_test.reset_index(drop=True).copy()
X_grouped.columns = [feature_map[col] for col in X_grouped.columns]
X_grouped = X_grouped.groupby(level=0, axis=1).mean()

# Get one scalar base value for class 1 (>50K) base_value =
explainer.expected_value
if isinstance(base_value, (list, np.ndarray)):
    base_value = base_value[1] base_value =
float(base_value)

# Waterfall plot shap_exp_grouped =
shap.Explanation(
    values=shap_grouped.iloc[pred_low].values,
    base_values=base_value,
    data=X_grouped.iloc[pred_low].values,
    feature_names=X_grouped.columns.tolist()
)

shap.plots.waterfall(shap_exp_grouped, max_display=10)

```

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